

Sound Biosynthetic Design over Iterative Grammars: A Realizability Geometry for Fungal Polyketides

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Abstract

[Stub — written last. Spine: wrong object (BGC $\rightarrow$ structure regression, the 51–68% ceiling) $\rightarrow$ program space $y = \text{Exec}_\Theta(z; G)$ run forward / inverse / backward $\rightarrow$ a sound triage tool with named limits. Most of the abstract is the framing-note one-paragraph thesis.]

1 Introduction

[Stub — drafted after §6 and §3. Wrong frame $\rightarrow$ right frame (program space; inference and design); the one-sentence thesis; architecture figure. ¶2 lever: Cox (2023) identifies machine learning as the route forward where rational HR-PKS engineering “will remain extremely difficult” — taken as a design constraint, not a validation. ¶3: structure elucidation here proceeds by expression + NMR with MS as triage; we build a triage tool grounded in metabologenomics observables and name explicitly where triage is insufficient.]

2 Substrate: a sound executor and verifier

[Stub — edited from the inverse-compiler manuscript (`paper/main.tex`). Sound deterministic executor + beam verifier; the three-state verdict (VERIFIED / UNDER-OBSERVED / OUT-OF-GRAMMAR); guarantees relative to operator library, grammar, observables, mass tolerance, adduct model.]

3 Characterizing the natural manifold

The natural fungal PKS manifold has **eight exactly-reachable cores**, a **0.57 control-axis ceiling with a structural mechanistic explanation**, and a **tailoring-permissive interior with 64% conditional reconstructibility** — this section characterizes its shape. We read these results not as a catalogue of where prediction fails but as the geometry the design contribution of §6 operates over: the boundary of what nature builds, the axis along which it is hard to read, and the interior where it is tractable once a core is in hand.

3.1 The boundary: eight exactly-reachable cores

Running the verifier inverse over all 326 fungal PKS entries in MIBiG 4.0, exact reconstruction reaches **8/326** under the implemented operator grammar. This is a lower bound — the missing

operators are enumerable (PT multi-ring aromatic closures, oxidative rearrangements), and adding coverage moves the number up — but the more important reading is that 8/326 is not a disappointing accuracy. It is the *boundary condition* of the natural manifold: the set of cores the current sound grammar can place exactly, the edge from which §6 measures design distance. “Only 8” is the wrong frame; “the manifold’s boundary is here, and it is sharp” is the right one.

3.2 Similarity is not a reconstructibility metric

A natural temptation is to soften the boundary with a structural-similarity proxy — count a core “reconstructed” if a predicted skeleton embeds in the product. We show this proxy is *unreliable*: it swings from 8 to 160 reconstructed cores as a single nuisance parameter (the allowed ring-closure slack) is varied, and it admits non-PKS false positives (a de-heteroatomated alkaloid skeleton can embed in a polyketide chain). Similarity is therefore not a reconstructibility metric; we keep exact match as the airtight signal and use the tailoring-latent reformulation below as the principled relaxation, not a tunable proxy.

3.3 The tailoring-permissive interior

Most MIBiG products are a PKS core plus post-PKS tailoring, so exact core-match understates reconstructibility. Moving tailoring into the latent — $z = (z_{\text{PKS}}, z_{\text{tail}})$ with a gene- and formula-budgeted edit applier — the joint search exactly reconstructs **7/106** annotated clusters overall, but the decisive number is conditional: of the 99 failures, **95 are PKS-core-unreachable** (no producible backbone), and among the clusters where a core *is* producible the tailoring search succeeds near-uniquely at **64%**. The bottleneck is PKS-core cyclization coverage, not tailoring. So the manifold’s interior is *tailoring-permissive*: once a core is in reach, the decorations are near-uniquely reconstructible, and the residual hardness is concentrated on the core, not the periphery.

3.4 The hard axis: the iteration-grammar wall, structurally explained

The manifold’s one genuinely hard axis is per-cycle control. A cross-taxa transfer probe makes it quantitative: the per-step reduction chemistry is learnable from abundant bacterial modular data and transfers to fungal cycles *perfectly* when each cycle’s active reductive domains are given (1.00), but collapses to **0.567** when only the synthase’s fixed domain set is given — the iteration-grammar wall. The wall decomposes into a narrow highly-reducing per-cycle-reduction control wall (~9% of products) and a broad, coverage-addressable aromatic-cyclization wall (~61%).

Why the wall is structural, not a data gap. A fungal iterative synthase reuses one set of catalytic domains across every cycle; the per-cycle outcome is fixed by the chain-length and substrate context the enzyme sees on that pass, not by any per-cycle change in the (constant) machinery. This is the precise form of Cox’s observation that highly-reducing PKS programming is an emergent property that “will remain extremely difficult” to predict rationally (Cox, 2023): it is emergent from the interaction between the constant catalytic machinery and the changing substrate context across cycles. The 0.57 is therefore not a placeholder for missing work; it is a structural ceiling for any predictor whose input is the synthase’s fixed domain content.

A sequence-level probe separates two questions. The cross-taxa probe originally folded together two questions we can now separate: does protein *sequence* carry per-step chemistry signal that the domain abstraction discards, and does that signal close the fungal wall? The first is testable

on bacterial *modular* PKS, where each module carries its own ketoreductase (KR) sequence: under homology-clean evaluation at 70% sequence identity,¹ a frozen ESM-2 head extracts inactive-domain and stereochemistry signal at balanced accuracy ~ 0.85 (AUC 0.91 for inactive-domain detection), decisively above the chance baselines of 0.50–0.53 that the domain cartoon is pinned to. The second question is answered structurally rather than empirically: a fungal iterative synthase reuses *one* KR sequence across all its cycles, so a per-domain sequence head emits one prediction per synthase — precisely the fixed-domain condition that yields 0.567. The per-cycle signal is not in the constant sequence; the wall is structural, not a data gap. Sequence carries per-step chemistry where there is a per-step sequence (bacterial, modular) and cannot where there is not (fungal, iterative).

This manifold — bounded by eight reachable cores, structured by the iteration-grammar wall, with a defined tailoring-permissive interior — is the object the design contribution of §6 operates over. Its hard axis is exactly the axis along which the interesting designs live (§6.6), which is why the same measurement reads as a prediction ceiling here and a design frontier there.

4 Generality across grammars

[Stub — ONE PAGE. PKS, NRPS, and a typed PKS–NRPS hybrid behind one interface; the load-bearing point is that the observability and verdict layers are family-agnostic, not the demo counts. Reviewers wanting more cross-grammar work are pointed at the repository.]

5 Mining: genome and metabolome

[Stub — ONE PAGE. The BGC $\leftrightarrow$ peak assignment graph; structure verdict $\perp$ attribution verdict; one real blind cluster $\rightarrow$ mass case (6-MSA, MIBiG BGC0001275). A bridge, not a pillar.]

6 Sound biosynthetic design over the iteration grammar

Running the executor forward turns a program into a structure; running the verifier inverse turns a structure into the set Z^* of programs that could have produced it. Here we run the grammar *backward*: given a desired, possibly non-natural target, we place it on a *realizability geometry* relative to the natural manifold, and we rank the experiments that would confirm it. Two results make this more than a restatement of the diagnostic half. First, the design space decomposes onto two axes that map to fundamentally different wet-lab capabilities. Second—and this is what keeps the contribution honest—the cost on those axes is the field’s *measured* difficulty of fungal PKS engineering, curated from primary literature, not an assumption we supply.

6.1 The realizability geometry: two axes

A design target is a program z_d ; its product is $y_d = \text{Exec}_\Theta(z_d; G)$. We score z_d by its edit distance to the nearest natural cluster on the manifold, decomposed into two typed axes:

¹Leave-cluster-out by sequence-identity cluster; identity computed via BLOSUM62 global alignment in Biopython, verified at self-identity 1.00 and an observed cross-cluster identity floor of 0.35. A strict 30%-identity partition is undefined here: KR is a single conserved Rossmann-fold family with a $\sim 35\%$ pairwise-identity floor, so it cannot be split into remote-homolog folds — the head’s generalization across remote KR homologs is therefore not assessable on this dataset, a statement about the protein family rather than the method.

- **Structural** edits change the *domain content*: add a reductive domain (`add_reductive`), swap the starter or extender (`starter`, `extender`), reprogram the release cyclase (`release`), or insert a methyltransferase de novo (`add_cmt`). These map onto bacterial modular-PKS engineering.
- **Control** edits change the *iteration program* with the domain set fixed: which cycle a present domain fires on (`reduction`, `c_methyl`) or the iteration count (`cycle_add`, `cycle_remove`). This is the iteration-grammar frontier—precisely the 100% $\rightarrow$ 57% wall of the cross-taxa transfer probe (§3), now read as a design dimension rather than a prediction limit.

A design at coordinate (2, 1) (two structural, one control edit) is a different proposition from one at (3, 0): the first requires reprogramming the iteration itself, the second is documented domain engineering. Each target’s most-realizable route is the minimum-cost edit path to a natural cluster, and its *verdict*—NATURAL, ENGINEERABLE, FRONTIER, or SPECULATIVE—is read off that path. Crucially, the control axis *only exists as a design dimension* for a machine that distinguishes per-cycle firing within a fixed domain set; a tool without the iteration grammar cannot represent the edit “reprogram the ketoreductase at cycle 3,” let alone cost it (§6.6).

6.2 A literature-grounded, super-additive cost

The design enumeration bounds the cost model to the nine edit kinds that actually appear in the 1-edit neighbourhood, and we curated each from primary sources rather than scraping abstracts (two tier rulings turned on facts a search snippet stated incorrectly). The structural edits inherit the mature combinatorial-biosynthesis canon: acyltransferase/extender swaps and loading-module/starter swaps are documented (AT-engineering review, 2018; Erythromycin analogues, 1997), as are reductive-loop transplants (Reductive-loop swaps, 2008); de novo C-methyltransferase insertion has no precedent (the methylation literature reprograms a *present* C-MeT, never inserts one) and is scored SPECULATIVE.

The control edits are the unsolved frontier. Cox’s review of iterative highly-reducing PKS programming (Cox, 2023) states plainly that rational efforts “will remain extremely difficult” because the programme is “an emergent property... highly unpredictable.” Yet a *single* edit beside a closely related cluster is achievable: domain swaps between the tenellin and desmethylbassianin synthases mapped methylation-pattern and chain-length changes cleanly (Fisch et al., 2011). We encode exactly this shape. Structural edits sum linearly by tier; control edits are **super-additive**—quadratic in the number k of control edits in a design:

$$\text{cost}(z_d) = \underbrace{\sum_{e \in \text{struct}} \tau(e)}_{\text{linear}} + \underbrace{\left(\sum_{e \in \text{ctrl}} \tau(e) \right) \cdot k_{\text{ctrl}}}_{\sim k^2, \text{ super-additive}} \quad (1)$$

where τ is the curated per-edit tier. Equation (1) reduces to the linear tier cost at $k = 1$ (the Fisch–Cox single-edit regime) and grows quadratically as edits stack (the Cox emergent regime). The cost model is therefore not asserted: it is the published difficulty landscape of fungal PKS engineering, written into the objective. This is the difference between “we built a design engine” and “we built a design engine whose cost model is the measured difficulty of the field.”

6.3 The worked neighbourhood of 6-MSA

Figure 1 walks one clean template—the 6-methylsalicylic acid synthase, the cluster with a verified blind real-mass match (§5)—end to end. Each row of its 1-edit neighbourhood is a build proposal carrying its (structural, control) coordinate, realizability verdict, predicted monoisotopic mass, and

[Figure: the 1-edit design neighbourhood of 6-MSA, as a ranked build table. Source: `scripts/worked_design_6msa.py`. Columns: (S,C) edit decomposition; verdict; predicted monoisotopic mass; verifying observable. Two labelled blocks (engineerable / frontier), executor-verifiable designs leading.]

Figure 1: The worked 6-MSA design neighbourhood: design distance and verification plan on one line.

the observable that would confirm it. The neighbourhood separates cleanly into an **engineerable** block (single structural edits—starter swaps such as acetyl→hexanoyl at mass 244.13, release reprograms—all confirmable by accurate mass) and a **frontier** block (single control edits such as reprogramming the cycle-3 reduction), with coverage-limited out-of-grammar designs sunk below the headline. One row reads, “one documented edit (acetyl→hexanoyl) from the 6-MSA synthase, mass 244.13, verifiable by accurate mass”; another, “one iteration-program edit (reprogram cycle-3 reduction), mass 172.07, needs MS/MS.” Design distance, the predicted observable, and a verdict on whether triage suffices or structure elucidation is required—on the same line.

6.4 Control-axis dominance

Across the 1-edit neighbourhood of the natural manifold the control axis dominates: **69** designs lie on the frontier (involve a control edit) against **49** engineerable (structural-only), with 24 speculative. That ratio is itself a product of the literature curation—the raw, uncured count was a starker 93:34, and grounding the edit costs (demoting release to a structural domain swap, raising de novo C-MeT and iteration-count edits) softened it to a defensible margin. Fungal biosynthetic *design* is mostly iteration-program editing: the axis only a sound per-cycle executor can express.

6.5 The experiment-selection planner

Given a design, which experiment should one run next? The planner ranks each modelled observation o by expected information gain per unit cost, weighted by realizability:

$$\text{score}(o) = \mathbb{E}[\Delta|Z^*|] \cdot w_{\text{real}}/c_{\text{obs}}(o), \quad w_{\text{real}} = 1/\text{cost}(\text{unresolved edits}), \quad (2)$$

with speculative designs hard-excluded ($w_{\text{real}} = 0$). The two cost terms sit on different sides: observation cost c_{obs} (accurate mass 1, MS/MS 3, in committed relative units) in the denominator, and the super-additive realizability cost of Eq. (1) as the design-side weight. Because that weight is super-additive, resolving one of two stacked control edits drops the residual penalty non-linearly—so the planner reasons about the *joint resolution state* of a design, not merely the most-uncertain edge. The planner’s output is a decision tree over observation outcomes in which both children of every node are first-class: the refuted branch (the observation is inconsistent with the predicted product) carries as much information as the confirmed one.

The planner returns a typed triage verdict. **VERIFIED-BY-TRIAGE**: the modelled observables collapse $|Z^*|$ to near-unique. **ELUCIDATION-REQUIRED**: the modelled observables are exhausted and $|Z^*|$ is still large—the design must go to structure elucidation. **OUT-OF-GRAMMAR**: a mass observation refutes every legal program. *Naming the elucidation-required case is the contribution*: a triage tool that knows its own limits, and whose limits are typed and machine-checkable. Figure 2 contrasts two trees: 6-MSA collapses at accurate mass (a shallow tree—the planner doing its job), while the highly-reducing C₁₂ core BAB stays under-determined through mass and MS/MS (4050→25→23 candidates) and is routed to expression + NMR (the planner naming its limit).

[Figure: two decision trees side by side. Source: `scripts/planner_tree_figure.py` → `results/planner_tree_{6msa,bab}.mmd`. Left: 6-MSA, root decision (*run mass?*) → *confirmed* ⇒ *verified-by-triage*; *refuted* ⇒ *out-of-grammar*. Right: BAB, *mass* → *MS/MS exhausted* ⇒ *elucidation-required*; *refuted* leaves drawn first-class at each level.]

Figure 2: The triage planner doing its job (6-MSA) and naming its own limit (BAB). Refuted branches are rendered first-class, not collapsed.

This scoping is forced by the data, not chosen for modesty. All seven verified products in our corpus had their genome→structure link established by heterologous expression (with knockout or native expression) and NMR, with accurate mass and MS/MS serving as triage and detection—never as the sole structure-determining observable. The planner is therefore a metabologenomics *triage* tool: it answers “which observation collapses $|Z^*|$ given the genome,” not “how is this structure proven.” Evaluated retrospectively against this curated corpus it routes correctly—the small aromatics (orsellinic acid, 6-MSA, mellein, 6-hydroxymellein) verify by accurate mass, while the highly-reducing core is flagged elucidation-required—and on the design neighbourhood it steers attention to engineerable and frontier designs over speculative ones.

6.6 The architectural moat and the same-wall symmetry

Two claims carry the contribution. **First, it is architectural, not a capability we happen to have.** The control axis exists as a design dimension only for a sound executor that distinguishes per-cycle firing within a fixed domain set. “Reprogram the ketoreductase at cycle 3” is not a hard edit to a retrosynthesis tool (which works in chemical space), an enzyme-design tool (which works per protein), or a pathway tool (which ignores assembly-line grammar)—it is a sentence about an object none of them can represent. We are the missing layer between generative chemistry and protein design, and the cost on the axis is the field’s measured difficulty (Eq. (1)). **Second, the same wall has opposite valence in the two directions.** The 100%→57% iteration-grammar collapse is, for *prediction*, a ceiling—where reconstruction fails (§3); for *design*, it is the frontier—the dimension along which the interesting molecules live. One measurement, a limitation read one way and the contribution read the other.

7 Limitations and outlook

[Stub — drafted after §6. *v0 NRPS/hybrid grammars; literature-curated edit tiers (the defensibility moat, curated from primary sources)*. Triage scoping, third placement: *the planner selects experiments and names elucidation-required clusters — it does not close the design–build–test loop*. Named *v1 extensions*: *observation-updated probabilistic realizability (tiers as success distributions, needing wet-lab build-outcome data)*; *the per-cluster mining planner*; and *bacterial generalization (ClusterCAD manifold density) as the post-paper companion piece, where it lands as confirmation rather than competition*.]

8 Conclusion

[Stub.]

References

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